OSPE DAY 4

SPOT NO. 1.

1. Write the names of GIT hormones A, B and C
   Ans: A → Cholecystokinin (CCK); B → Secretin; C → Gastrin
2. Write down the functions of any two hormones as chyme enters the GI tract.
   Ans: CCK: 1. Contraction of Gallbladder; 2. Relaxation of sphincter of Oddi; 3. Inhibits gastric emptying; 4. Stimulates acinar cells of pancreas to release a juice rich in pancreatic digestive enzymes
   Secretin: 1. Inhibits stomach motility and secretion. 2. Inc. in pancreatic & biliary HCO3 secretion
   Gastrin: 1. Stimulates parietal cells to secrete acid; 2. Stimulate chief cells to secrete pepsinogen; 3. Inc. contraction of stomach and relaxation of pylorus helping in emptying; 4. Inc. LES tone; 5. Relaxes ileocecal valve; 6. Contributes to gastrocolic reflex.

SPOT NO. 2.

1. Name the type of contractions seen in the picture.
   Ans: Haustral movements
2. What is the purpose of these contractions?
   Ans: Mechanical mixing of food to allow for increased absorption. Some forward movement also takes place.
3. How much water is absorbed by large intestine and how much is passed in feces?
   Ans: Large intestine absorbs 1.4 L; 100 ml passes in feces.

SPOT NO. 3.

1. What are the phases of swallowing?
   Ans: Oral phase, Pharyngeal phase, Esophageal phase.
2. The bolus of food touches which areas in oral cavity & pharynx to initiate the swallowing reflex?
   Ans: Tonsillar pillars, tonsils, soft palate, base of the tongue, posterior pharyngeal wall
3. Write any two factors that prevent food from going down the larynx
   Ans: 1. Larynx elevates and causes epiglottis to close the glottis; 2. Vocal folds adduct; 3. Deglutition apnea. 4. Cough reflex

SPOT NO. 4.

1. Which process is shown in the picture?
   Ans: Enterohepatic circulation
2. How much bile salts are reabsorbed by this process and from where?
   Ans: 95% bile salts are reabsorbed; site is terminal ileum.
3. What else is absorbed from this portion of small intestine?
   Ans: Vitamin B_{12}

SPOT NO. 5.

1. Name the transporters A & B by which iron is absorbed from intestine.
   Ans: A → Heme transporter; B → Divalent Metalo Transporter – 1 (DMT1)
2. Name the transporter C on the basolateral side by which iron is released
   Ans: Ferroportin
3. What is the role of Hepcidin?
   Ans: Hepcidin prevents the release of iron from Ferroportin.